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Q3: Component Theory and Principles of Operation

1. Rain Sensor Module (Resistive Type)

- **Theory & Principle of Operation:** The rain sensor works on the principle of electrical conductivity. It consists of a plate with exposed nickel-coated tracks that act as a variable resistor. When the plate is dry, the resistance between the tracks is extremely high (theoretically infinite), resulting in an open circuit. When rain falls on the sensor, water acts as a conductor, bridging the gap between tracks and completing the circuit. This lowers the resistance significantly, allowing current to flow.
- **Interfacing:** The module typically has a Digital Output (DO) and an Analog Output (AO). For this project, the **Analog Output (AO)** is interfaced with the Arduino's Analog-to-Digital Converter (ADC) pin (e.g., A0). This allows the microcontroller to measure the intensity of the rain based on the varying voltage drop, rather than just a simple ON/OFF state.

2. Microcontroller (Arduino UNO / ATmega Family)

- **Theory:** The system utilizes a microcontroller from the **ATmega family** (specifically the ATmega328P on the Arduino UNO). It serves as the central processing unit.
- **Principle of Operation:** The microcontroller operates by executing a program written in **C language**. It continuously polls the analog signal from the rain sensor. When the sensor signal crosses a predefined threshold (indicating rain), the microcontroller processes this data and generates a specific control signal to the actuator.
- **Interfacing:**
 - **Input:** Receives analog voltage signals from the Rain Sensor on pin **A0**.
 - **Output:** Sends Pulse Width Modulated (PWM) signals from a digital PWM pin (e.g., Pin 9) to the servo motor signal wire.
 - **Power:** Provides regulated 5V power to both the sensor and the servo motor.

3. Servo Motor (Actuator)

- **Theory & Components:** The servo motor is a rotary actuator that allows for precise control of angular position. It consists of four main parts: a DC motor, a gear system, a position sensor (mostly a **potentiometer**), and control electronics.
- **Principle of Operation (PWM Control):** The servo is controlled using **Pulse Width Modulation (PWM)** signals. It expects a pulse every **20 milliseconds** (50 Hz refresh rate). The width of the pulse determines the angular position of the shaft:
 - A pulse width of **1 millisecond** moves the servo to **0 degrees**.
 - A pulse width of **1.5 milliseconds** moves the servo to **90 degrees**.
 - A pulse width of **2 milliseconds** moves the servo to **180 degrees**.
- **Feedback Mechanism:** The DC motor is connected to a gear mechanism that drives the output shaft and simultaneously turns the internal potentiometer. As the potentiometer rotates, its resistance changes, producing a voltage corresponding to the current position. The internal error amplifier compares this voltage to the input PWM signal voltage. If there is a difference (error), the motor spins until the error is zero, ensuring accurate positioning.
- **Interfacing:** The servo has three wires:
 - **VCC (Red):** Connected to 5V.
 - **GND (Brown/Black):** Connected to Ground.
 - **Signal (Orange/Yellow):** Connected to the Arduino PWM pin (Pin 9).